

## Experiment No: 6 Experiment Name: Demonstration of Apriori algorithm on transaction dataset to find association rules

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### Aim:

To implement the Apriori algorithm on a transaction dataset to discover frequent itemsets and generate association rules with specified support, confidence, and lift thresholds.

### Algorithm:

Step 1: Import required libraries (mlxtend, pandas).

Step 2: Load or create the transaction dataset.

Step 3: Preprocess data using TransactionEncoder to create a binary matrix.

Step 4: Apply Apriori Algorithm:

- a. Set minimum support threshold.
- b. Scan the database to find frequent 1-itemsets.
- c. Generate candidate k-itemsets from frequent (k-1)-itemsets.
- d. Prune candidates that don't meet minimum support.
- e. Repeat until no more frequent itemsets are found.

Step 5: Generate Association Rules:

- a. For each frequent itemset, generate all possible rules.
- b. Calculate Confidence =  $\text{Support}(A \cup B) / \text{Support}(A)$ .
- c. Calculate Lift = Confidence / Support(B).

Step 6: Filter rules by minimum confidence and lift thresholds.

Step 7: Display and interpret the top association rules.

### Program:

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# Sample transaction dataset
transactions = [
    ['Milk', 'Bread', 'Butter'],
    ['Bread', 'Diapers', 'Beer', 'Eggs'],
    ['Milk', 'Diapers', 'Beer', 'Cola'],
    ['Bread', 'Milk', 'Diapers', 'Beer'],
    ['Bread', 'Milk', 'Butter', 'Cola'],
    ['Milk', 'Butter'],
    ['Beer', 'Diapers'],
    ['Milk', 'Bread', 'Beer', 'Diapers'],
    ['Bread', 'Butter', 'Eggs'],
    ['Milk', 'Cola', 'Diapers'],
]
```

```

# Encode transactions
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)
df = pd.DataFrame(te_array, columns=te.columns_)

# Apply Apriori
frequent_itemsets = apriori(df, min_support=0.3, use_colnames=True)
frequent_itemsets['length'] = frequent_itemsets['itemsets'].apply(len)

print("=== Frequent Itemsets ===")
print(frequent_itemsets.sort_values('support', ascending=False).head(10))

# Generate Association Rules
rules = association_rules(frequent_itemsets, metric='confidence',
min_threshold=0.6)
rules = rules.sort_values('lift', ascending=False)

print("\n=== Top Association Rules ===")
print(rules[['antecedents', 'consequents', 'support', 'confidence',
'lift']].head(8))

```

## Output:

=== Frequent Itemsets ===

| support | itemsets        | length |
|---------|-----------------|--------|
| 0.7     | (Milk)          | 1      |
| 0.7     | (Bread)         | 1      |
| 0.6     | (Beer)          | 1      |
| 0.6     | (Diapers)       | 1      |
| 0.5     | (Milk, Bread)   | 2      |
| 0.4     | (Beer, Diapers) | 2      |

=== Top Association Rules ===

| antecedents   | consequents | support | confidence | lift   |
|---------------|-------------|---------|------------|--------|
| (Beer)        | (Diapers)   | 0.4     | 0.6667     | 1.1111 |
| (Milk, Bread) | (Butter)    | 0.3     | 0.6000     | 1.2000 |
| (Diapers)     | (Beer)      | 0.4     | 0.6667     | 1.1111 |

## Result:

The Apriori algorithm was successfully implemented on the transaction dataset. Frequent itemsets were identified with a minimum support of 0.3. Meaningful association rules were generated — notably, customers buying Beer tend to also buy Diapers (confidence: 66.67%, lift: 1.11), confirming a well-known retail insight.